

PRODUCT REQUIREMENTS DOCUMENT

Rental Housing Law Navigator

Which rules apply here today, and what is about to change?

VERSION	1.0 (draft for build)
DATE	October 3, 2026
PREPARED BY	Sayan Shil
COMPANION	Technical Requirements Document (TRD) v1.0
SCOPE	3 states, 10 cities, 6 rule categories

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1. Document Overview

Product	Rental Housing Law Navigator (RHLN)
Event	Hack-Nation 7th Global AI Hackathon, Challenge 02, powered by RealPage
Document	Product Requirements Document (PRD)
Version	1.0 (draft for build)
Date	October 3, 2026
Author	Sayan Shil
Companion document	Technical Requirements Document (TRD) v1.0
Source material	Challenge brief (6 pages). The starter pack and participant guide were not available when this was written.

1.1 Purpose

This document defines what we are building, for whom, and how we will know it works. It lists every feature, rule and deliverable for the 24-hour build. The TRD explains how to build it.

1.2 How to read this document

- **Priority tags.** P0 means required to compete. P1 means should have. P2 means nice to have.
- **[Confirm]** marks items that depend on the starter pack or participant guide. Check them in the first hour.
- **Status words.** The five result statuses are applies, unknown, superseded, not yet effective and pending. They are used the same way in every section.
- **Scope.** All laws named here come from the challenge brief. The brief says the answer key has not been reviewed by counsel. Nothing in this product is legal advice.

Before you build

Hour 0 to 1 task: open the starter pack and compare the rule schema, the answer key format and the participant guide with the assumptions listed in Section 19. Update this PRD if anything differs.

2. Executive Summary

Rental housing in the United States is regulated in layers. State statutes, county rules and city ordinances each have their own coverage tests, effective dates and exemptions. The answer to "what rules apply to this apartment?" depends on the exact address, and it changes often.

RHLN is an AI system that reads public housing law and answers one question for any apartment address: **which rules apply here today, and what is about to change?** Every answer is cited, dated and honest about what it does not know.

2.1 What we are building

Module	Name	What it does	Output
A	Extract	Reads the 87-document law corpus with an AI agent and writes one structured record per rule, each with a verified quote.	rules.json
B	Resolve and Apply	Geocodes an address, builds its state, county and city stack, and tests each rule against building facts.	lookups.json
C	Track change	Runs the six change tests, supports "as of" queries, and separates enacted from pending law.	changes.json
Explain	Plain language	Shows each rule in simple words with a citation. English first, Spanish as a stretch goal.	Web app

2.2 Key product decisions

1. **The AI reads. Code decides.** The model turns legal text into structured data. A deterministic engine decides which rules apply to an address. This keeps answers repeatable and auditable.
2. **"Unknown" is a first-class answer.** When a rule depends on a fact we do not have, we say unknown. We never guess. The brief confirms that unknown earns partial credit.
3. **No quote, no rule.** A rule enters the dataset only if its quoted source text is found word for word in the corpus.
4. **Every answer has an as-of date.** Enacted law and pending law are always shown apart.
5. **One pipeline for everything.** The same automated pipeline handles the 87 starter documents and the ordinance released at hour 16. Nothing is hand-coded.

2.3 What success looks like

- A judge types an address and sees the jurisdiction stack, every applicable rule, and a source quote for each, in under 30 seconds.
- The team shows score.py output on the dev set, results for tests T1 to T6, and the hour-16 ordinance processed live.
- Unknown answers explain which fact is missing.
- Adding a new jurisdiction means adding documents, not writing code.

3. Problem and Opportunity

The source text of housing law is public. But it is unstructured, scattered and constantly changing. Getting a reliable answer today means reading municipal codes, statutes and pending bills by hand.

Signal from the brief	Why it matters to the product
14 cities and counties across 8 states have enacted local bans on algorithmic rent-setting since late 2024, most with different definitions and penalties.	Local rules differ in small ways. A tool must read each one, not assume they match.
January 2026: California amended its antitrust law on common pricing algorithms (AB 325 / SB 763). It added a state layer on top of local rules.	The same address can be covered by both a state and a local rule, with different effective dates.
June 2026: Massachusetts' high court removed a statewide rent-control question from the November ballot.	Proposed changes do not always become law. The tool must know the difference and must not report rules that do not exist.

3.1 Who feels the pain

- **Renters** do not know their rights at their own address: rent increase limits, deposit caps, fee rules, eviction protections.
- **Small housing providers** do not have legal teams. They do not know their obligations before they act.
- **Advocates and agencies** cannot see the full picture across jurisdictions, or what a pending bill would change.

3.2 Why it is hard

Difficulty	Example from the corpus
Layered law	California has a statewide rent cap and just-cause rule, plus local rent control in cities such as San Francisco and Los Angeles.
Coverage tests	Rules apply by building age, unit count or owner type. The SF Rent Ordinance applies when the certificate of occupancy is on or before 6/13/1979.
Precedence	The AB 1482 state cap yields to the SF Rent Ordinance. Massachusetts law bars local rent control. The NJ FAIR Act may preempt two local bans.
Time	Laws take effect later (NJ FAIR Act: signed 7/20/2026, effective 7/1/2027). Bills stay pending. Measures get struck.
Missing data	Parcel data has no year built for San Diego, and no year built or units for Berkeley. No owner names exist at all.

3.3 Opportunity

Modern language models can read long legal text and return structured data. The hard part is trust. RHLN is designed so that every claim can be traced to a quoted sentence, every uncertainty is shown, and every change is dated. That is the intended outcome in the brief: make housing law visible at the level of a single address.

4. Goals, Non-Goals and Success Criteria

4.1 Goals

ID	Goal
G1	Extract rules automatically from the 87-document corpus into schema-valid records, each with a verified quote.
G2	Resolve every sample address to a jurisdiction stack using legal city limits, not postal city.
G3	Return every applicable rule for an address with a plain-language summary and a citation.
G4	Say "unknown" whenever coverage depends on a fact the data does not contain.
G5	Show what changes by date: "as of" queries, enacted versus pending, and before and after rule sets.
G6	Show conflicts and preemption honestly, including possible conflicts.
G7	Extend to a new jurisdiction by adding documents only.

4.2 Non-goals

ID	We will not
N1	Give legal advice or a compliance certification.
N2	Suggest ways to avoid, structure around or evade a rule.
N3	Cover states beyond California, New Jersey and Massachusetts (except the live new-jurisdiction demo).
N4	Cover federal law, lease review, court cases or individual tenancy facts such as current rent.
N5	Use customer, resident, pricing or any other non-public data.
N6	Scrape sites against their terms. We use the starter corpus and free public sources.
N7	Build user accounts, payments or notifications.
N8	Predict whether a pending bill will pass.

4.3 Success criteria mapped to scoring

The brief scores 100 points. Seventy-five are automatic. Teams self-test with score.py on the dev key. The targets below are our own internal goals.

Component	Points	How it is measured	Our target
Extraction accuracy	25 auto	Rules matched to the held-out key by jurisdiction, category and citation. Field accuracy on date, status, key value, citation.	Match every dev-key rule. All fields correct on matched rules.
Address coverage	20 auto	100 held-out addresses. Missing an applicable rule costs double. Unknown earns partial credit.	Zero missed applicable rules on the 20 dev addresses. Unknown only when a fact is truly missing.
Citations	15 auto	Share of "applies" answers backed by a source and a quoted span found in the corpus.	100% of results carry a verified quote.
Change tracking	15 auto	Overlap with expected affected-address sets for T1 to T6, plus conflict flags on T3.	All six tests pass. T3 conflict flagged.
Plain language and usability	10 judges	Demo.	A non-expert understands a result in 30 seconds.
Responsible design	10 judges	Uncertainty, audit trail, guardrails.	Each guardrail is visible in the demo.
Scalability path	5 judges	How the approach extends to new jurisdictions.	Live add of a new ordinance with no code change.

5. Users and Personas

Persona	Context and need	Key questions	Success moment
Renter	Lives in a multifamily building. Wants to know rights at this address. Not a lawyer. May read English or Spanish.	Can my rent go up this much? Is there a cap on my deposit? Can I be evicted without a reason? What are the fees?	Reads a short answer with a source and knows what to do next.
Small housing provider	Owns a few buildings. No legal team. Wants to understand obligations before acting.	Which rules cover my building? Does a small-owner exemption apply? What changes next year?	Sees a clear list, sees where the answer is unknown, and sees what fact would settle it.
Advocate or agency analyst	Works across many addresses and cities. Needs the full picture.	Which buildings does this bill affect? Which cities have the strongest protections? Where do state and local rules conflict?	Runs a change test and gets a list of affected addresses with before and after rules.
Judge or reviewer (stakeholder)	Scores the demo and the outputs. Time is short.	Is it accurate? Is it cited? Does it say unknown? Is it safe?	Sees scores, citations, guardrails and a live rerun in the demo.

5.1 User stories

ID	As a...	I want to...	So that...	Pri
US-01	Renter	enter my address and see the rules that apply today	I know my rights.	P0
US-02	Renter	see a source quote and link for each rule	I can check it myself.	P0
US-03	Renter	read each rule in plain language	I can act on it.	P0
US-04	Renter	switch the view to Spanish	I can read it in my language.	P1
US-05	Provider	see which of my obligations are certain and which are unknown	I know what to check.	P0
US-06	Provider	add facts I know, such as unit count or year built	unknown results can be settled.	P1
US-07	Provider	see rules that take effect soon	I can plan ahead.	P0
US-08	Advocate	run a change case and list affected addresses	I see who a law change touches.	P0
US-09	Advocate	see the rule set before and after a date	I understand what changes.	P0
US-10	Advocate	see pending bills clearly marked as proposed	I do not confuse them with law.	P0
US-11	Advocate	see where a state and a local rule conflict or may conflict	I can flag it for review.	P0
US-12	Analyst	query any address "as of" a past or future date	I can audit history.	P0
US-13	Analyst	drop in a new ordinance and see new rules and affected addresses	I can react to a new law fast.	P0
US-14	Reviewer	see an audit trail of sources, model outputs and changes	I can trust the system.	P1
US-15	Reviewer	see a confidence score and a conflict flag on every answer	I know when to review by hand.	P1

6. Scope

The brief sets the scope at 3 states, 10 cities and 6 rule categories. It is small enough to finish and varied enough to be hard.

6.1 Jurisdictions

State	Cities in scope	Why it is in the set
California	Los Angeles, San Francisco, San Diego, Berkeley, Santa Ana*	Densest layering: statewide rent cap and just cause, local rent control, state and local algorithmic-pricing rules with different effective dates.
New Jersey	Jersey City, Hoboken, Newark	Rent control is set city by city. Statewide eviction, screening and fee rules. Two local algorithmic-pricing bans and a new statewide FAIR Act (effective July 2027) that may preempt them.
Massachusetts	Boston, Cambridge	State law bars local rent control. A 2026 ballot question was struck before the vote. Algorithmic-pricing bills are pending. Tests whether systems avoid reporting rules that do not exist.

*Santa Ana laws are in the corpus for extraction. No open parcel data with addresses exists for Santa Ana, so the address sample covers the other 9 cities.

6.2 Rule categories

#	Category	What to capture	Examples named in the brief
1	Rent increase limits	Cap formula, covered buildings, exemptions, local versus state precedence	CA Tenant Protection Act, Civ. Code 1947.12 (5% + CPI, max 10%). SF Rent Ordinance, Admin. Code ch. 37. LA Rent Stabilization Ordinance. MA G.L. c.40P (state bar on local rent control).
2	Just-cause eviction	Allowed causes, notice, relocation assistance, coverage	CA Civ. Code 1946.2. NJ Anti-Eviction Act, N.J.S.A. 2A:18-61.1.
3	Security deposits	Maximum, exceptions, effective date	CA Civ. Code 1950.5 as amended by AB 12 (one month; two for qualifying small landlords; effective 7/1/2024). NJ N.J.S.A. 46:8-21.2 (1.5 months). MA G.L. c.186 15B (first month's rent).
4	Application and screening fees	Fee caps, allowed upfront charges, receipts and refunds	CA Civ. Code 1950.6 (CPI-adjusted cap). NJ P.L.2025, c.405 (\$50 cap, effective 5/1/2026). MA G.L. c.186 15B. MA broker-fee rule, G.L. c.112 87DDD1/2 (8/1/2025).
5	Screening restrictions	Limits on criminal-history and income-source screening; timing rules	NJ Fair Chance in Housing Act (2021). CA source-of-income protections under FEHA (SB 329).
6	Algorithmic rent-setting	Definition of covered software, prohibited conduct, penalties, effective date	CA AB 325 / SB 763 (1/1/2026). San Francisco 37.10C (Oct 2024). San Diego 98.1101 to 98.1104 (Jun 2025). Berkeley ch. 13.63 (2026). Santa Ana Ord. NS-3090 (Apr 2026). Jersey City 218-12 (Jun 2025). Hoboken ch. 158, Art. II (Jul 2025). NJ FAIR Act, P.L.2026, c.43 (7/1/2027). MA S.2983 / H.5222 (pending).

The answer key holds 58 rules and 19 "no rule at this level" findings. 52 of the 58 rules were verified against public sources as of October 1, 2026. It has not been reviewed by counsel.

6.3 Facts we have and facts we lack

The sample is about 500 multifamily properties in 9 cities from public assessor data. Each has street, postal city, ZIP, year built, units and use code. There are no owner names. Teams resolve the legal jurisdiction themselves.

Fact	Availability	Effect on results
Year built	Available except San Diego and Berkeley	Age-based coverage becomes unknown where missing. Edge years are unknown.
Unit count	Available except Berkeley	Unit-based coverage becomes unknown where missing.
Use code	Available (varies by county)	Normalized to a property type.
Owner type and portfolio size	Never available	Small-owner exemptions are unknown unless the user supplies the fact.
Certificate of occupancy date	Not available (year built is a proxy)	Dates inside a year are unknown. Example: a 1979 building and the 6/13/1979 test.
Current rent, tenancy length, tenant details	Not available and not wanted	Rules that need them cannot be fully resolved. We say unknown.

6.4 Out of scope

Everything in the non-goals list (Section 4.2), plus: federal fair housing law, eviction court procedure, local registration or licensing rules not in the six categories, and law-firm or news pages. The brief provides law-firm and news pages as links only. We treat them as non-authoritative and do not extract rules from them.

7. Product Principles and Responsible AI

The brief asks for transparency, not legal verdicts. These principles are requirements, not preferences. Each maps to features in Section 8.

The solution should	The solution must not
Cite the source text and retrieval date for every rule it reports.	Present output as legal advice or a compliance certification.
Show an "as of" date on every answer and separate enacted from pending law.	Suggest ways to avoid, structure around or evade a rule.
Say "unknown" when coverage depends on facts it does not have.	Invent rules or citations where the source text is silent.
Flag conflicts and low-confidence answers for human review.	Use customer, resident, pricing or other non-public data.
Explain rules in plain language that a renter can act on.	Scrape sites in violation of their terms of use.
Keep an auditable log of sources, model outputs and changes.	

7.1 Status vocabulary

These five statuses are the only result values in lookups.json. The labels below are what people see in the app.

Status	English label	Spanish label	Meaning
applies	Applies	Aplica	In force on the as-of date and the building meets all conditions.
unknown	Unknown (facts missing)	No se sabe (faltan datos)	In force, but a needed fact is missing. Or precedence depends on a missing fact.
superseded	Overridden by another rule	Reemplazada por otra norma	Another rule takes precedence at this address. Both are shown.
not yet effective	Not yet in effect	Aún no está vigente	Enacted, with an effective date after the as-of date.
pending	Proposed. Not law.	Propuesta. No es ley.	A bill or proposal that has not been enacted. Never shown as in force.

7.2 Guardrails at a glance

- "Not legal advice" appears in every interface, every export and every video.
- No result is shown without a verified quote, a source link and a retrieval date.
- Unknown results say which fact is missing. They do not suggest how to change the outcome.
- Struck, withdrawn or repealed measures are never reported as rules.
- A possible state and local conflict is flagged, not hidden.
- Every model call, source and change is written to an audit log.

8. Functional Requirements

Every feature has an ID, a priority and an acceptance test. IDs are used in the TRD and in the build plan. P0 features are required for the minimum viable entry and for the three required modules.

8.1 Module A: Rule extraction

ID	Feature	Description	Pri	Acceptance criteria
A-01	Corpus ingestion	Load the 87 documents and the manifest CSV (URL, retrieval date, type). Mark link-only pages as non-authoritative.	P0	All 87 loaded. Count matches manifest. Link-only items listed, not extracted.
A-02	Document profiling	Classify each document (statute, ordinance, bill, regulation). Find jurisdiction, short title and enacted or pending signals.	P0	Every document has jurisdiction and type. Spot-check of 10 documents passes.
A-03	Section-aware chunking	Split by section markers. Attach definitions and exemption sections as context.	P0	No chunk loses its heading. Definitions are visible to the extractor.
A-04	Structured rule extraction	The model returns one record per rule: category, jurisdiction, requirement, coverage, exemptions, effective date, status, penalty, citation, quoted span.	P0	Output validates against the schema 100%. Dev-key score reported.
A-05	Quote verification	Each quote must match the document text exactly after normalizing spaces and quote marks. Store character offsets.	P0	Zero records without a verified quote in rules.json.
A-06	Coverage and exemption conditions	Convert coverage text into machine-checkable conditions on units, year built, owner type, property type and location.	P0	Conditions parse and evaluate. Missing facts give unknown.
A-07	Dates and status	Extract effective date (absolute or relative) and status (enacted or pending). Compute relative dates when the base date is stated.	P0	Date and status match the dev key. Relative dates computed or flagged.
A-08	Penalty and key values	Capture penalties, caps, formulas and amounts (for example 5% + CPI, max 10%).	P0	Key value present whenever the text states one.
A-09	Deduplication and merge	Merge duplicates across chunks and documents with a stable key. Link amendments to the base rule.	P0	No duplicate rule IDs. Amendments linked.
A-10	Relation extraction	Extract precedence statements: yields to, preempts, supplements, amends.	P0	SF over AB 1482 and the MA c.40P bar are found from text.
A-11	Non-rule event registry	Record struck, withdrawn or failed measures (for example the MA ballot question) so they are never reported as rules.	P0	T5 shows no rent cap for Boston or Cambridge.
A-12	Plain-language summaries	One short summary per rule, grounded in the quote. English first. Spanish as a stretch.	P0 / P1	No new facts in the summary. Reading level about grade 8.
A-13	Resumable, cached runs	Cache by document hash and prompt version. Resume after a failure.	P0	A rerun makes no new model calls.
A-14	Dev-key self-score	One command runs score.py on the dev key and prints the report.	P0	Report saved with a run ID.
A-15	One-command ingest of a new document	Add a document and get new rules, relations and affected addresses.	P0	Hour-16 ordinance processed end to end in under 5 minutes.
A-16	Confidence and review queue	Score each record. Queue low-confidence items. Reviewers can flag or annotate but not edit extracted fields.	P1	Queue visible in the console. Extracted fields are read-only.
A-17	Multi-format input	Accept txt, md, html, pdf and docx for new documents.	P1	Each format yields text with stable offsets.

8.2 Module B: Address lookup (resolve and apply)

ID	Feature	Description	Pri	Acceptance criteria
B-01	Address intake	Single-address form. CSV batch with street, postal city, ZIP, year built, units, use code.	P0	All ~500 sample rows load. Bad rows are listed.
B-02	Normalization and geocoding	Use the Census batch geocoder. Cache results. Store match type.	P0	Geocode coverage reported. Cache hit on rerun.
B-03	Legal city resolution	Decide the city from the incorporated place, not the postal city.	P0	No Hoboken address is treated as Newark, and the reverse.
B-04	Jurisdiction stack	Return state, county and city. San Francisco is one city and county.	P0	Stack correct for all sample cities.
B-05	Fallback resolution	If the geocoder fails, use point-in-polygon on TIGER boundaries. Mark lower confidence.	P1	No address ends without a stack or a clear reason.
B-06	Fact normalization	Normalize year built, units and use code into a property type. Keep the source of each fact.	P0	Every fact has provenance.
B-07	Data gap handling	Treat San Diego (no year built), Berkeley (no year built or units) and owner type as unknown.	P0	Affected rules return unknown, not applies or not covered.
B-08	Candidate rule selection	Collect all rules in the stack for the six categories.	P0	Candidate list complete for a test address.
B-09	Lifecycle by as-of date	Decide in force, not yet effective, pending or ended.	P0	Boundary dates pass (see T1 and T3).
B-10	Coverage test	Evaluate coverage and exemptions with true, false and unknown logic.	P0	Unknown propagates correctly. Unit tests pass.
B-11	Precedence	Apply extracted relations. Mark superseded rules and show which rule overrides which.	P0	SF 20-unit example matches the brief mock-up.
B-12	No-rule findings	For each category and level, say "No rule found in the provided corpus at this level" when none exists.	P0	Findings match the key's "no rule" entries on the dev set.
B-13	Result statuses	Return applies, unknown, superseded, not yet effective or pending per rule.	P0	Only these five values appear.
B-14	Citations	Each result has citation label, URL, retrieval date and verified quote.	P0	Quote found in the source text at stored offsets.
B-15	Missing-fact guidance	For unknown results, list the missing fact and where a person may find it. No advice on changing the outcome.	P0	Every unknown names at least one missing fact or reason.
B-16	Batch run and export	Compute all ~500 addresses and export lookups.json.	P0	File validates. Runs without errors.
B-17	Confidence and conflict flag	Score each answer. Flag conflicts and low confidence.	P1	Flag shown in the UI and in exports.
B-18	User-supplied facts	Let a person add facts they know, labelled "user-supplied". Used for that session only.	P1	Result changes only through the supplied fact. Label shown.
B-19	Upcoming and pending panel	List not-yet-effective and pending rules for the address.	P0	Panel shows dates and a "Proposed. Not law." label for pending items.

8.3 Module C: Change tracking

ID	Feature	Description	Pri	Acceptance criteria
C-01	Temporal rule model	Each rule has effective date, end date and lifecycle (enacted, pending, struck, withdrawn, repealed).	P0	Lifecycle drives status at any date.

ID	Feature	Description	Pri	Acceptance criteria
C-02	As-of query	Every lookup and rule query accepts an as-of date. Default is Oct 1, 2026 [Confirm].	P0	Same address gives different results across dates.
C-03	Change case runner	Run T1 to T6 and write changes.json.	P0	All six produce expected affected sets on self-check.
C-04	Before and after diff	For each affected address list the rule set before, after, and the difference.	P0	Diff shows added, removed and changed items.
C-05	Pending-bill impact	Evaluate pending bills as a hypothetical. Never report them as in force.	P0	T4 lists addresses, status stays pending.
C-06	Conflict flags	Flag possible preemption or overlap between state and local rules.	P0	T3 flags the NJ FAIR Act against the Hoboken and Jersey City bans.
C-07	Live ordinance handling	New document to new rules to affected addresses, with the future effective date right.	P0	T6 passes with no code change.
C-08	Address timeline	Show how one address's rules change over a date range.	P1	Timeline lists each change date.
C-09	Ad hoc impact analysis	Pick a rule or document and see affected addresses.	P1	Works for any rule in the dataset.
C-10	Change export	Write changes.json with affected addresses and conflict flags for each test.	P0	Structure matches the submission format [Confirm].

8.4 Presentation and user experience

ID	Feature	Description	Pri	Acceptance criteria
D-01	Address search	Search box with sample-address suggestions and a free-text option.	P0	A result appears in under 3 seconds.
D-02	Result by category	Six category cards with status badge, summary and citation chip.	P0	All six categories always shown, even when "no rule".
D-03	Jurisdiction stack view	Breadcrumb of state, county, city with rule counts per level.	P0	Shown on every result.
D-04	Source viewer	Side drawer with source text and the quote highlighted.	P0	Highlight matches the stored offsets.
D-05	As-of control	Date picker and quick jumps (today, effective dates).	P0	Changing the date refreshes statuses.
D-06	Upcoming and pending panel	See B-19.	P0	See B-19.
D-07	Unknown explanation	A box that says why a result is unknown and what is missing.	P0	Present on every unknown.
D-08	Disclaimer banner	"Not legal advice" on every page, export and video.	P0	Cannot be dismissed.
D-09	Spanish view	Toggle for English and Spanish on all result text.	P1	Switch needs no reload.
D-10	Change explorer	List of T1 to T6 with before and after diff and an affected-address table.	P0	Counts match changes.json.
D-11	Pipeline console	Run extraction, upload a document, watch progress, see failures.	P1	Live progress for a run.
D-12	Audit trail viewer	Search sources, model calls and changes.	P1	Each result links to its audit events.
D-13	Score report viewer	Show the latest score.py report.	P1	Matches the file output.
D-14	Accessibility and mobile	Keyboard use, readable contrast, screen-reader labels, mobile layout.	P1	No blocking issues on a quick audit.

ID	Feature	Description	Pri	Acceptance criteria
D-15	Result export	Download one result as JSON or a printable page.	P2	Export carries the disclaimer and as-of date.

8.5 Responsible AI controls

ID	Feature	Description	Pri	Acceptance criteria
R-01	Citation gate	No rule or result without a verified quote, source and retrieval date.	P0	Automated check on every export.
R-02	Unknown policy	Missing facts give unknown. Omission and guessing are both disallowed.	P0	Unit tests on all dev-set gaps.
R-03	Enacted versus pending	Separate in data, labels and layout.	P0	T4 and T5 pass.
R-04	No-evasion filter	Summaries and guidance must not suggest ways to avoid or structure around a rule.	P0	Red-team prompts produce refusal text. Filter tests pass.
R-05	No invented law	Absence is stated as "no rule found in the provided corpus".	P0	Struck ballot question never appears as a rule.
R-06	Conflict and low-confidence flags	Show flags and queue items for human review.	P1	Flagged items listed.
R-07	Audit log	Append-only log of sources, model calls, outputs, changes and exports.	P1	Every lookup links to its audit trail.
R-08	Public data only	Use only the starter pack and free public sources. No scraping against terms.	P0	Source list in the README.
R-09	Privacy	No owner names or personal data stored. User-supplied facts are not saved by default.	P0	Database review shows no personal data.

8.6 Stretch goals

ID	Feature	Description	Pri	Acceptance criteria
S-01	Spanish renter view	Plain-language Spanish for every rule and status label.	P1	All statuses and summaries available in Spanish.
S-02	Confidence and conflict flag	Per-answer score and flag, see B-17.	P1	Visible in UI and exports.
S-03	Live new-jurisdiction demo	Add one new jurisdiction during the event by adding documents only.	P1	New city rules appear with no code change.

8.7 Submission features

ID	Feature	Description	Pri	Acceptance criteria
Z-01	rules.json	All rule records in the provided format, with citation and quoted text.	P0	Validates against the schema.
Z-02	lookups.json	For all 500 addresses, each rule's result status.	P0	All 500 present. Only the five statuses.
Z-03	changes.json	Affected addresses and conflict flags for each test.	P0	T1 to T6 present.
Z-04	GitHub repository	Code, README with run steps, and the three output files.	P0	Fresh clone runs by following the README.
Z-05	Live demo link	A working link to the tool.	P0	Loads and answers a lookup.
Z-06	Three videos	Team, demo and technical videos. They include the scores.	P0	Each video meets the rules in Section 17.

9. Business Rules

These rules decide what the product shows. They are written so they can be tested.

ID	Rule
BR-01	As-of is inclusive. A rule with effective date D is in force on D and after.
BR-02	Enacted with a future effective date is "not yet effective". Enacted with an effective date on or before the as-of date is in force. Pending is never in force.
BR-03	Pending items are shown apart from law and labelled "Proposed. Not law."
BR-04	Struck, withdrawn or repealed measures never appear as rules. They appear only in change history.
BR-05	The legal city (incorporated place) decides which city rules apply. The postal city does not.
BR-06	A rule covers an address only if every coverage condition is true and no exemption is true. If no condition is false and at least one is unknown, the result is unknown.
BR-07	If any coverage condition is false, the rule is left out of results. It stays visible in the trace on request.
BR-08	If a local rule overrides a state rule, the state rule is "superseded" and both citations are shown. If the override depends on an unknown fact, both rules are "unknown".
BR-09	If rules supplement each other (for example state and local algorithmic-pricing rules), both apply.
BR-10	Missing a rule that applies is worse than saying unknown. When unsure, use unknown.
BR-11	Absence is stated as "No rule found in the provided corpus at this level." We never say no law exists.
BR-12	A rule without a verified quote cannot enter the dataset.
BR-13	A year-built value that equals the year of a date threshold is unknown. Years before are before. Years after are after.
BR-14	Relative effective dates (for example "60 days after adoption") are computed only when the base date is stated. Otherwise the date is empty and the result is unknown.
BR-15	A possible conflict is flagged when a future-effective state rule overlaps a local rule in the same category and no savings clause is found.
BR-16	User-supplied facts override dataset facts for that session only and are labelled.
BR-17	"Not legal advice" appears in every interface and export.
BR-18	The product never suggests how to restructure, time or describe a property to avoid a rule.

10. User Journeys and Screens

10.1 Journeys

J1. Renter checks an address

1. Opens the app. Sees the disclaimer and a search box.
2. Types an address. Picks the as-of date or keeps the default.
3. Sees the jurisdiction stack and six category cards.
4. Reads a plain-language line for each rule. Opens a source drawer to read the exact text.
5. Sees an "Upcoming and pending" panel for rules that start soon or are only proposed.

J2. Provider meets an unknown

1. Looks up a building. Sees "Unknown (facts missing)" on a small-landlord exemption.
2. Reads which fact is missing (owner type) and why it matters.
3. Optionally adds a fact they know. The result updates and is labelled "user-supplied".

J3. Advocate checks a pending bill (T4)

1. Opens the change explorer. Picks MA S.2983 / H.5222.
2. Sees the label "Proposed. Not law." and the list of Massachusetts addresses it would touch.
3. Sees that no address shows the bill as "applies".

J4. Analyst processes the hour-16 ordinance (T6)

1. Opens the pipeline console. Uploads the new ordinance file.
2. Watches extraction, quote verification and relation steps finish.
3. Opens the change case. Sees the new rule, its future effective date and affected Cambridge addresses.
4. Runs a lookup "as of" before and after the effective date to confirm the status change.

J5. Team adds a new jurisdiction (stretch)

1. Registers the jurisdiction (name, level, FIPS or place code).
2. Uploads its documents. The same pipeline runs.
3. Looks up an address in the new city. Rules appear with citations. No code changed.

10.2 Screen inventory

Screen	Route	Purpose	Key elements	Pri
Home / lookup	/	Start a lookup	Search, as-of, language, disclaimer, sample addresses	P0
Result	/lookup	Show rules for an address	Stack, summary bar, six category cards, upcoming panel, conflict banner	P0
Source drawer	overlay	Show the exact source text	Highlighted quote, link, retrieval date	P0
Rule explorer	/rules	Browse all rules	Filters, status, source link	P1
Rule detail	/rules/[id]	Inspect one rule	Fields, conditions, relations, timeline	P1
Change explorer	/changes	List T1 to T6	Case cards, status, counts	P0
Change case	/changes/[id]	Before and after	Diff, affected table, conflict flags	P0
Address timeline	/timeline	History by date	Date line of rule changes	P1
Pipeline console	/pipeline	Run and watch extraction	Runs, live log, upload, review queue	P1

Screen	Route	Purpose	Key elements	Pri
Audit trail	/audit	Trace sources and calls	Search, filters, linked events	P1
Score report	/eval	Show scores	Dev-set report, test results	P1
About and limits	/about	Explain method and limits	Disclaimer, data sources, known gaps	P0

10.3 Result screen wireframe

Not legal advice. Summaries of public law. Always check the source text.

Rental Housing Law Navigator

EN | ES

100 Example Street, San Francisco, CA 94110

As of: 2026-10-01

Facts: 1962 | 20 units

Search

California

>

City & County of San Francisco

4 apply

1 unknown

2 upcoming

Rent increases

Summary + citation chip

APPLIES

Just cause

Summary + citation chip

APPLIES

Security deposit

Summary + citation chip

APPLIES

Screening fee

Summary + citation chip

UNKNOWN

Algorithmic pricing

Summary + citation chip

APPLIES

Source drawer

Cal. Civ. Code 1950.5

Retrieved: [date] | Link

Quoted span highlighted here in the original source text

Why this applies

units = 20 (assessor)

year built = 1962 (assessor)

owner type = unknown

Conflict flag: none

Upcoming and pending

Rule X: not yet in effect, starts [date] | Bill Y: Proposed. Not law.

Schematic only. The example uses the brief's sample: a 20-unit building in San Francisco built in 1962. The screening-fee status is shown as unknown to illustrate the pattern.

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11. Change-Tracking Test Cases

These six tests come from the brief. Expected affected addresses are held by the judges, so we also write our own checks from the rule text.

Test	Scenario	What a correct system does	How we check
T1	CA AB 325 / SB 763, effective 1/1/2026	"Not yet effective" for CA addresses as of 12/31/2025. "Applies" as of 1/2/2026.	Run lookups on both dates. Also check 1/1/2026 (inclusive).
T2	Hoboken and Jersey City local bans	Each ban applies only inside its own city limits. Neither applies in Newark.	Use incorporated place. Check borders and postal-city mismatches.
T3	NJ FAIR Act, signed 7/20/2026, effective 7/1/2027	"Not yet effective" today. "Applies" on 7/2/2027. Flags a possible conflict with the two local bans.	Run lookups at 3 dates. Check the conflict flag.
T4	MA S.2983 and H.5222 (pending bills)	Reports them as pending, never in force. Lists the addresses they would affect.	No "applies" status for these rules at any date.
T5	MA rent-control ballot question, struck 6/23/2026	Reports no rent cap for Boston or Cambridge. The affected set is empty.	Rule list for MA rent contains no cap. Affected set equals zero.
T6	Fictional Cambridge ordinance (released at hour 16)	Extracts it unaided, lists affected addresses, gets the future effective date right.	Run the pipeline on the file. Check rule, date and affected list.

11.1 Extra regression checks

- SF 20-unit building built in 1962: the SF Rent Ordinance applies and the AB 1482 cap is superseded (matches the brief's mock-up).
- A 1979 building in San Francisco: the certificate-of-occupancy test is unknown.
- A Berkeley address with no year built and no units: age and size rules return unknown.
- A Massachusetts address: "no rule" for local rent control, and the c.40P state bar is shown.
- Any lookup with a rule whose quote cannot be verified: the rule never reaches the output.

12. Deliverables and Output Specifications

Three files are scored. Final field names must match the provided schema [Confirm]. Our internal format may carry extra fields. An export step maps them.

File	Contents	Validation
rules.json	One record per rule: category, jurisdiction, requirement, coverage conditions, exemptions, effective date, status (enacted or pending), penalty, source citation, quoted span.	Schema check. Every quote found in the corpus. No duplicates. No struck or withdrawn items.
lookups.json	For all 500 addresses, each rule's result: applies, unknown, superseded, not yet effective or pending. Includes jurisdiction stack and as-of date.	All 500 present. Only five statuses. Each "applies" has a quote.
changes.json	For each test T1 to T6: affected addresses, before and after rule sets, conflict flags (T3).	All six tests present. Counts match the change engine.

12.1 Other deliverables

- **GitHub repository** with code, a README that explains how to run it, and the three output files.
- **Live demo link** to a working tool.
- **Team video**: introduce the team.
- **Demo video**: show the tool in use.
- **Technical video**: explain how the system works.
- Each video must show the full score.py report on the dev set, results for T1 to T6, and the system processing the hour-16 ordinance.

13. Non-Functional Requirements

Area	Requirement	Target
Performance	Single lookup from precomputed rules	p95 under 1 second; under 3 seconds for a cold computation
Performance	Batch lookup for 500 addresses	Under 2 minutes after geocoding is cached
Performance	Hour-16 ordinance end to end	Under 5 minutes, one document
Performance	Full corpus extraction	Under 30 minutes with parallel calls; free on rerun (cache)
Reliability	Demo availability	Static fallback with cached results if a service is down
Accuracy	Citations	100% of reported rules carry a verified quote
Accuracy	Dates	Boundary-date tests pass (before, on, after)
Determinism	Same inputs and cache give the same outputs	Model temperature 0. Prompt version logged.
Security	Write endpoints protected	API key. Rate limits. Input validation.
Privacy	No personal data	No owner names. No tenant data. User facts not stored by default.
Accessibility	Basic WCAG-style checks	Keyboard use, contrast, labels
Localization	English and Spanish	Statuses and summaries
Auditability	Trace every answer	Source, quote, model call, run ID
Maintainability	New jurisdiction by data only	No code change for a new city
Cost	Stay inside credits	Cache everything. Use a cheaper model for simple steps.
Compliance	Source terms	Starter corpus and free public sources only

14. Data Requirements

Source	What it provides	Use in product	Notes
Starter law corpus	87 documents (statute, ordinance and bill text) plus manifest CSV with URL and retrieval date. Law-firm and news pages are links only.	Module A input	Official text only. Retrieval date shown to users.
Sample addresses	~500 multifamily properties in 9 cities. Street, postal city, ZIP, year built, units, use code.	Module B input	No owner names.
Rule schema	JSON Schema with required fields and a worked sample record.	Output format	Map internal fields to it.
Dev answer key	10 rules and expected results for 20 addresses.	Self-testing	Run often.
Change test cases	The 6 tests. Expected sets held by judges.	Module C	
Scoring script	score.py, the same script judges use.	Self-scoring	Show output in videos.
Census Geocoder	Address to coordinates, state, county, place.	Module B	No key. Batch up to 10,000 addresses.
Census TIGER/Line	City and county boundary shapefiles.	Fallback and checks	Free download.
Parcel and assessor data	Year built, units, use code (MassGIS, NJ MOD-IV, LA County, SF).	Facts	Gaps: San Diego year built; Berkeley year built and units; Santa Ana none.
LegiScan and Open States	Bill text, status, votes.	Optional pending-bill checks	Free keys. LegiScan: 10,000 queries a month.
LSC Eviction Laws Database	Coded eviction laws as of 1/1/2021.	Methods only	Not current truth.

Data boundaries

Not provided and not allowed: customer or resident data, rent or pricing data, internal legal analysis, and scraping that breaks a site's terms. Municipal code sites often restrict bulk scraping. Use the starter corpus.

15. Metrics and Evaluation

15.1 Internal quality metrics

Metric	Definition	Goal
Dev-key extraction score	score.py extraction component on the dev key	Top of range; no regressions
Quote verification rate	Records whose quote is found exactly / records produced	100% in rules.json
Missed-applicable-rule rate	Applicable rules answered as omitted / all applicable rules	0 on the dev set
Unknown rate	Unknown results / all results, with a reason for each	Explained, not hidden
Citation coverage	Applies results with a verified quote / applies results	100%
Change-test pass rate	T1 to T6 matched to our own expectation	6 of 6
Hour-16 time	Minutes from file received to affected list	Under 5
Cost per full run	Model spend for corpus extraction	Within credits, with room for 3 reruns

15.2 Evaluation loop

1. Run extraction on a few documents. Read the output by eye.
2. Run the dev-key score. Record the report.
3. Change the prompt or code. Rerun only what changed (cache).
4. Repeat until the score stops improving. Freeze the prompt version.
5. Run the full corpus. Run lookups for 500 addresses. Run T1 to T6.
6. Run score.py one last time and capture the report for the videos.

16. Release Plan and Milestones

The hackathon plan in the brief sets the rhythm. Our build milestones follow it.

Hours	Milestone	Exit criteria
0 to 1	Kickoff and setup	Starter pack read. Schema and dev key understood. Repo created. Assumptions confirmed.
1 to 6	Module A running	All 87 documents processed. Dev-key score recorded. Quote gate on.
6 to 11	Module B core	Geocoding cached. Stacks correct. Coverage engine passes unit tests.
11 to 16	Address lookup complete	lookups.json for 500 addresses. Citations. English summaries. UI result page.
16 to 20	Change tracking	Hour-16 ordinance processed. T1 to T6 pass. Conflict flag on T3.
20 to 23	Scoring and polish	score.py run. Fixes. Spanish view. Videos recorded.
23 to 24	Demo and submit	Files, repo, live link and videos submitted.

16.1 Cut line

If time runs short, drop work in this order. Never cut P0.

1. Result export (D-15) and map views.
2. Address timeline (C-08) and ad hoc analysis (C-09).
3. Audit viewer and score viewer screens (D-12, D-13). Keep the audit data.
4. Pipeline console UI (D-11). Keep the CLI.
5. Spanish view (S-01).
6. Confidence score formula detail (keep a simple version and the conflict flag).

16.2 Feature freeze

Freeze new features at hour 20. After that, only fixes, scoring runs, documentation and videos.

17. Demo and Submission Plan

17.1 Demo script (about 4 minutes)

Time	Scene	What to show	What it proves
0:00	Hook	The question. One address typed in.	Clear purpose
0:30	Result	SF 20-unit building. Stack, six categories, citations.	Accuracy, plain language
1:15	Source	Open the drawer. Highlighted quote and retrieval date.	Traceability
1:45	Unknown	A Berkeley or San Diego address with missing facts.	Honest uncertainty
2:15	As-of	Move the date across 1/1/2026 and 7/1/2027.	Change tracking
2:45	Changes	T1 to T6 and the T3 conflict flag.	Module C
3:15	Hour-16	Process the new ordinance live.	Automation, scalability
3:45	Scores	score.py report on screen.	Measured results

17.2 Video requirements (from the brief)

- **Team video:** introduce the team.
- **Demo video:** show the tool in use.
- **Technical video:** walk through how the system works.
- All videos must include test scores. Run score.py on the dev set and show the full report on screen. Show results for T1 to T6. Show the system processing the hour-16 ordinance dataset.
- Every interface shown must say "not legal advice".

17.3 Submission checklist

Item	Done when
rules.json	Validated. Quotes verified. In repo.
lookups.json	500 addresses. Only five statuses. In repo.
changes.json	T1 to T6. Conflict flags on T3. In repo.
README	Install, env vars, run steps, data sources, limits, disclaimer.
Live demo	Link opens. Sample lookup works. Disclaimer visible.
Videos	Three videos with scores and T1 to T6 and hour-16 shown.
Final score run	Report saved and shown in a video.

18. Risks and Mitigations

Risk	Impact	Mitigation
Model invents text or citations	Lost trust; citation points lost	Exact quote check. Reject on failure. Retry once. Log it.
Over- or under-extraction of rules	Extraction score falls	Clear prompt on "one record per rule". Dev-key loop. Merge duplicates.
Wrong dates or relative dates	Wrong status at as-of dates	Parse dates in code. Boundary tests. Flag relative dates.
Postal city differs from legal city	Wrong local rules	Use incorporated place from Census. Test border cases.
Missing parcel facts	Wrong "applies" or "not covered"	Unknown policy. Explain the gap.
Preemption is nuanced	Wrong precedence	Extract from text. Flag uncertainty. Show both citations.
Hour-16 document is a new format or style	T6 fails	Multi-format loader. Rehearse with our own synthetic ordinance.
Schema or matching differs from our assumption	Rework at the end	Read schema in hour 0 to 1. Export adapter. Run score.py early.
Rate limits or model outage	Extraction stalls	Caching. Provider fallback. Concurrency limit. Resume.
Credits run out	Cannot rerun	Estimate tokens first. Cheaper model for simple steps. Cache.
Live demo fails	Lost demo	Static fallback. Recorded backup. Precomputed lookups.
Answer key is not counsel-reviewed (6 of 58 rules unverified)	Some expected answers may be off	Trust the source text. Note limits in the README.
Time overrun	Missing P0	Cut line in Section 16.1. Freeze at hour 20.
Scope creep on UI	Less time for accuracy	Accuracy first. Keep the UI simple and clear.

19. Assumptions, Dependencies and Open Questions

19.1 Assumptions

- The rule schema has the fields listed in the brief: category, jurisdiction, requirement, coverage conditions, exemptions, effective date, status, penalty, source citation, quoted span.
- The default as-of date for scoring is October 1, 2026, as in the brief's example [Confirm].
- lookups.json lists rules whose result is one of the five statuses. Rules that clearly do not cover a property are left out [Confirm].
- The hour-16 ordinance arrives as text, markdown, HTML, PDF or DOCX in the shared folder.
- Using hosted language models and the sponsor credits is allowed.

19.2 Dependencies

- Starter pack in the shared Drive folder.
- Language-model API credits (Anthropic credits provided; other providers as backup).
- Census Geocoder and TIGER files (free, public).
- Hosting for a live demo (frontend, API, database).

19.3 Open questions

#	Question	Where to find the answer
1	What are the exact JSON field names and enums in the rule schema?	Starter pack, schema file
2	How does matching and partial credit work for statuses and key values?	Participant guide
3	Should lookups.json include rules that do not cover a property?	Participant guide, dev key
4	How does the key represent "no rule at this level" for counties?	Dev answer key
5	Does "superseded" need the superseding rule ID?	Schema and dev key
6	In what format will the hour-16 ordinance arrive?	Organizers
7	Is the team solo or a group? How should workstreams be split?	Team
8	Are time limits set for the three videos?	Organizers

Appendix A. Glossary

Term	Meaning
As-of date	The date for which rules are evaluated. Inclusive.
Coverage conditions	Tests that decide if a rule applies to a building: units, year built, owner type and similar.
Exemption	A condition that removes a building from a rule.
Jurisdiction stack	The state, county and city that govern an address.
Legal city	The incorporated place from the Census, not the postal city.
Preemption	When a higher level of law blocks or overrides a lower level.
Supersede	When one rule takes precedence over another at an address.
Enacted	Passed into law. May still have a future effective date.
Pending	Proposed, not law.
Not yet effective	Enacted, with an effective date after the as-of date.
Unknown	A rule is in force but a needed fact is missing.
No-rule finding	A statement that the provided corpus holds no rule for a category at a level.
Quoted span	The exact sentence or phrase from the source that supports a rule.
Starter pack	The data and code provided by the organizers.
Dev key	The small answer key for self-testing: 10 rules and 20 addresses.

Appendix B. Standard Text

Disclaimer (English)

This tool summarizes public law for information only. It is not legal advice and not a compliance certification. Laws change. Check the source text, or ask a qualified professional, before you act.

Disclaimer (Spanish)

Esta herramienta resume leyes públicas solo con fines informativos. No es asesoría legal ni una certificación de cumplimiento. Las leyes cambian. Revise el texto de la fuente o consulte a un profesional calificado antes de actuar.

Standard messages

Situation	Message
No rule at a level	No rule found in the provided corpus at this level.
Unknown	We cannot tell if this rule covers the building. Missing fact: [fact].
Superseded	This rule is overridden at this address by [rule]. Both are shown.
Not yet effective	This rule is enacted but starts on [date].
Pending	Proposed. Not law. This bill has not been enacted.
Possible conflict	A state rule and a local rule may overlap. Flagged for review.
Low confidence	Low confidence. Please check the source text.

Appendix C. Traceability to Scoring

Scoring component	Main features	Related business rules
Extraction accuracy (25)	A-01 to A-15	BR-12, BR-14
Address coverage (20)	B-01 to B-16	BR-05 to BR-10, BR-13
Citations (15)	A-05, B-14, R-01	BR-12
Change tracking (15)	C-01 to C-07, C-10	BR-01 to BR-04, BR-15
Plain language and usability (10)	A-12, D-01 to D-10, S-01	BR-17
Responsible design (10)	R-01 to R-09, B-17, D-07, D-08	BR-10, BR-11, BR-17, BR-18
Scalability path (5)	A-15, S-03, J5	None (data-driven design)